

Hsin Ya Lin

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WORK EXPERIENCE

Satellite Electronics Intern | Taiwan Space Agency | Jul 2026 to Aug 2026

- Designed the satellite reaction wheel test board in KiCad and built it, integrating an ESP32-S3 microcontroller, a 9 axis IMU over I2C, an RS422 transceiver and a DC-DC buck converter with hardware and software safety interlocks.
- Wrote roughly 9,000 lines of real time C and C++ firmware for that board, and cut combined communication and motor latency from 727ms to 263ms.
- Verified the RS422 link byte by byte by decoding captured wire traffic against an independent implementation, establishing 7 protocol facts that had been assumed and diagnosing 4 firmware defects including memory corruption behind a reboot loop.
- Ran a six stage gated bring-up and flew auto generated control code on target through a real IMU and four reaction wheels across 65 test flights on an air bearing stage.

RESEARCH

Digital Design and Verification | Purdue SoCET Matrix Coprocessor Team | Jan 2026 to May 2026

- Designed a matrix math coprocessor in SystemVerilog running on FPGA hardware, with a 33 state main control FSM, nested FSMs inside the arithmetic modules, and a memory map split three ways so operands and results cannot overwrite one another, verified with 7 module level testbenches.
- Carried the same RTL through an ASIC flow covering logic synthesis in Yosys, place and route, timing closure and standard cell layout in Cadence Virtuoso, with simulation in Icarus Verilog and waveform analysis in QuestaSim.

TECHNICAL EXPERIENCE

Power and Hardware Engineer | Purdue Lunabotics | Sep 2025 to Aug 2026

- Designed a 24V, 20A six cell battery sensing and protection board around the TI BQ77915 in KiCad, sizing the external resistor network that sets the overcurrent and overvoltage thresholds, and simulated the MOSFET switching and RC values in LTspice with a simplified model of the protector.
- Laid the board out for fabrication, hand assembled and micro soldered 50+ components onto it, diagnosed voltage and thermal variances during bench test, and integrated it into the rover power distribution system.

Power Electronics Design Projects | Purdue ECE 20007 | Jan 2026 to May 2026

- Designed and built an adjustable DC-DC boost converter stepping a 5V input to a 5V to 30V range, sizing the switching duty cycle from the boost transfer function rather than tuning it empirically, and added a PMOS output ceiling limiter that holds the rail under no load and under a shorted load.
- Designed a three band analog audio equalizer with active RC filters, op amp buffers placed to remove loading between the filter and volume stages, and a power amplifier stage driving over 400mW into a speaker.

EDUCATION

Purdue University | West Lafayette, Indiana | Aug 2025 to Present

Bachelor of Science in Electrical Engineering (expected December 2028)

GPA: 3.98 / 4.0; Dean's List (2 semesters) & Semester Honors (2 semesters)

SKILLS

- Languages: C, C++, MATLAB, Python, SystemVerilog, Assembly, shell scripting, Linux/Unix, Git
- Hardware: schematic through PCB layout and assembly in KiCad and Altium, micro soldering, surface mount rework, harness crimping, DC-DC buck and boost conversion, battery sensing and protection, active RC filters and op amp stages
- Embedded: bare metal C and C++ real time firmware, microcontroller bring-up, ESP32-S3 and MSP430, peripheral and sensor drivers, safety interlocks, SPI, I2C, UART, RS422, CAN bus, byte level protocol verification against captured wire traffic
- Digital: RTL design and verification in SystemVerilog, FSM and datapath, module level testbenches, memory mapped IO, RISC-V, hardware accelerator design, FPGA prototyping, logic synthesis through standard cell layout, RTL to GDS flow
- Systems and control: attitude determination and control, quaternion control, state estimation, Kalman filtering, 6 DoF rigid body dynamics, model based design with generated code verified against its model, six stage gated bring-up, hardware in the loop testing, parameter identification from captured data, oscilloscope, logic analyzer, LTspice